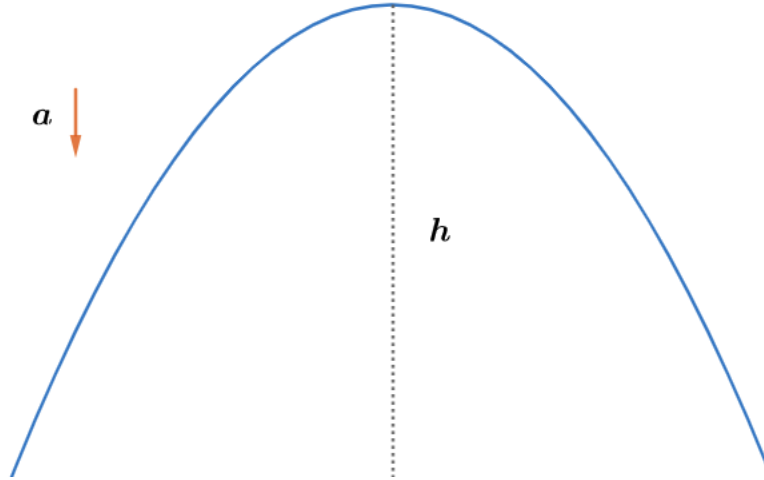


2021 F=ma Exam: Problem 9

Kevin S. Huang



From kinematics, we have

$$h = \frac{1}{2}a \left(\frac{T}{2} \right)^2$$

where a is the acceleration of the plane and T is the length of time on the parabolic arc. For the same change in altitude h , we have

$$T^2 \propto \frac{1}{a}$$

Using the fact that effective gravity $g_{\text{eff}} = g - a$, we see that to simulate weightlessness, $a = g$, while to simulate Mars, $a' = g - g'$:

$$\frac{T'}{T} = \sqrt{\frac{a}{a'}} = \sqrt{\frac{g}{g - g'}} = \sqrt{\frac{10 \text{ m/s}^2}{10 \text{ m/s}^2 - 3.7 \text{ m/s}^2}} = 1.26$$
$$T' = 1.26T = 31 \text{ s}$$

so the answer is B.